

# Intermediate Macro

## Chapter 19: Output, the Interest Rate, and the Exchange Rate

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# Table of Contents

- ① Chapter 19: Output, the Interest Rate, and the Exchange Rate
  - 19.1 Equilibrium in the Goods Market
  - 19.2 Equilibrium in Financial Markets
  - 19.3 Putting Goods and Financial Markets Together
  - 19.4 The Effects of Policy in an Open Economy
  - 19.5 Fixed Exchange Rates



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# Key Tools and Concepts

- Chapter 19 introduces a modern version of the **Mundell-Fleming** model, which is the  $IS - LM$  model in an open economy under perfect capital mobility. The treatment in the text differs from the canonical model by assuming that the expected future exchange rate is fixed, rather than equal to the current exchange rate.
- A **devaluation** is a decrease in the level of a fixed exchange rate (akin to a depreciation). A **revaluation** is an increase in the level of a fixed exchange rate (akin to an appreciation).

# Assumptions

- The domestic and foreign price levels are assumed to be fixed. Accordingly, expected domestic inflation and expected foreign inflation are assumed to be *zero*, which implies that the real interest rate equals the nominal interest rate throughout the world.
- Production is assumed to respond to changes in demand without changes in price, so demand determines output.
- Foreign currency is assumed to have no transactions value for domestic residents.
- Domestic and foreign bonds are considered perfect substitutes, and there is perfect capital mobility, so uncovered interest parity holds.
- The expected future exchange rate is assumed to be constant.

# Equilibrium Condition in the Goods Market Revisited

## 19.1 Equilibrium in the Goods Market

- Recall that we derived the equilibrium condition in the goods market in Chapter 18 (equation (18.4)):

$$Y = C(Y - T) + I(Y, r) + G + X(Y^*, \varepsilon) - IM(Y, \varepsilon)/\varepsilon$$

( + )
(+, -)
( + , -)
(+, +)

- It will be convenient in what follows to regroup the last two terms under net exports  $NX$ :

$$NX(Y, Y^*, \varepsilon) \equiv X(Y^*, \varepsilon) - IM(Y, \varepsilon)/\varepsilon$$

- Using this definition of net exports, we can rewrite the equilibrium condition as

$$Y = C(Y - T) + I(Y, r) + G + NX(Y, Y^*, \varepsilon) \quad (19.1)$$

( + )
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## 19.1 Equilibrium in the Goods Market

- ①  $P = P^* = 1$ , so  $\varepsilon = E$

Given our focus on the short run, we assumed that domestic and foreign price level are fixed and equal to one.

- ②  $\pi^e = 0$ , so  $i = r$

Since  $P$  is fixed, expected inflation is zero.

$$Y = C(Y - T) + I(Y, i) + G + NX(Y, Y^*, E) \quad (19.2)$$

# Domestic Bonds versus Foreign Bonds

## 19.2 Equilibrium in Financial Markets

- Under the assumptions of perfect asset substitutability (i.e., no risk premium) and perfect capital mobility, the expected return on domestic and foreign bonds must be same.
- Recall the uncovered interest parity condition derived in Chapter 17 (equation (17.2)):

$$(1 + i_t) = (1 + i_t^*) \left( \frac{E_t}{E_{t+1}^e} \right) \quad (19.3)$$

- Multiply both sides by  $E_{t+1}^e$  and reorganize to get

$$E_t = \frac{1 + i_t}{1 + i_t^*} E_{t+1}^e \quad (19.4)$$

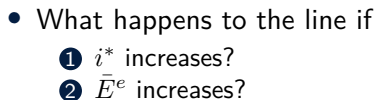
# Assumption on the Expected Future Exchange Rate

## 19.2 Equilibrium in Financial Markets

- For now, we shall take the expected future exchange rate as given and denote it as  $\bar{E}^e$  (we shall relax this assumption in Chapter 20).
- Under this assumption, and dropping time indexes, the interest parity condition becomes

$$E = \frac{1 + i}{1 + i^*} \bar{E}^e \quad (19.5)$$

## 19.2 Equilibrium in Financial Markets



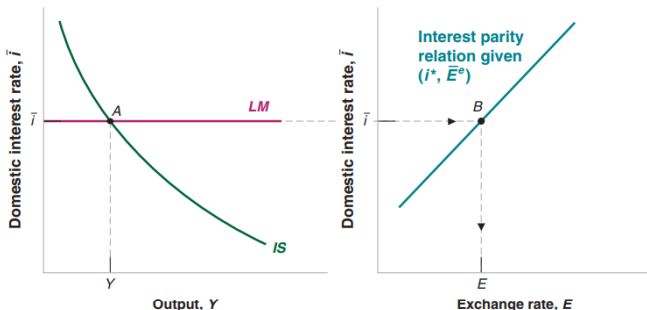
# The $IS - LM$ Model in an Open Economy

## 19.3 Putting Goods and Financial Markets Together

- The open economy versions of the  $IS$  and  $LM$  relations:

$$IS : Y = C(Y - T) + I(Y, i) + G + NX \left( Y, Y^*, \frac{1 + i_t}{1 + i_t^*} \bar{E}^e \right)$$

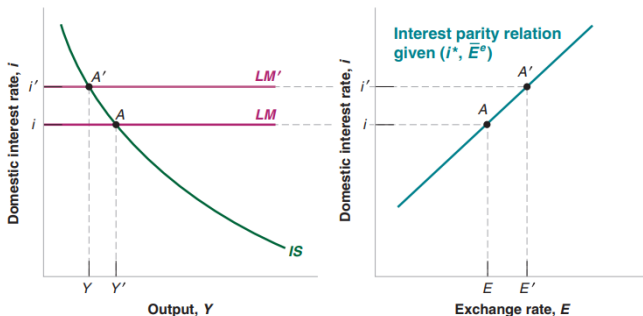
$$LM : i = \bar{i}$$



# The Effects of Monetary Policy in an Open Economy

## 19.4 The Effects of Policy in an Open Economy

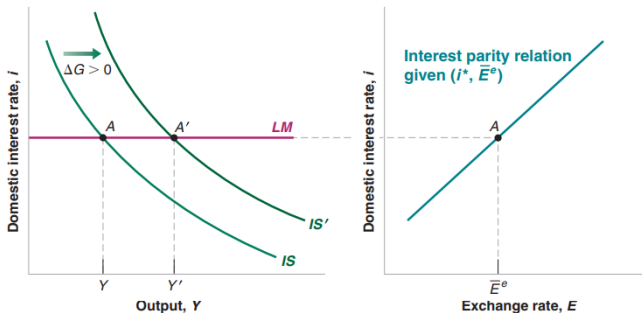
- Let us start from the effects of the central bank's decision to increase the domestic interest rate.



# The Effects of Fiscal Policy in an Open Economy

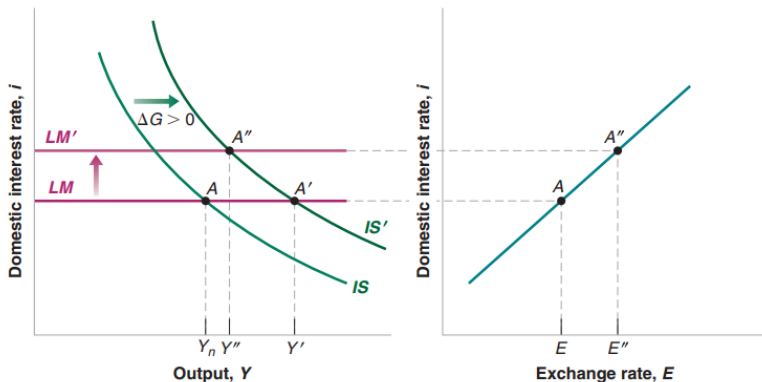
## 19.4 The Effects of Policy in an Open Economy

- Now let us look at change in government spending.
- Suppose that, starting from a balanced budget, the government decides to increase defense spending without raising taxes and so runs a budget deficit.



# When the Central Bank Responds by Raising the Interest Rate

## 19.4 The Effects of Policy in an Open Economy



# Monetary Contraction and Fiscal Expansion: The United States in the Early 1980s

## 19.4 The Effects of Policy in an Open Economy

- The combined effects of higher interest rates and a fiscal expansion were very much in line with what the Mundell-Fleming model predicts.
- By the mid-1980s, the **twin-deficits** (the budget deficit and the trade deficit) were the main macroeconomic policy issue.

**Table 1** Major US Macroeconomic Variables, 1980–1984

|                           | 1980 | 1981 | 1982 | 1983 | 1984 |
|---------------------------|------|------|------|------|------|
| GDP growth (%)            | −0.5 | 1.8  | −2.2 | 3.9  | 6.2  |
| Unemployment rate (%)     | 7.1  | 7.6  | 9.7  | 9.6  | 7.5  |
| Inflation (CPI) (%)       | 12.5 | 8.9  | 3.8  | 3.8  | 3.9  |
| Nominal interest rate (%) | 11.5 | 14.0 | 10.6 | 8.6  | 9.6  |
| Real interest rate (%)    | 2.5  | 4.9  | 6.0  | 5.1  | 5.9  |
| Real exchange rate        | 85   | 101  | 111  | 117  | 129  |
| Trade surplus (% of GDP)  | −0.5 | −0.4 | −0.6 | −1.5 | −2.7 |

Inflation: rate of change of the CPI. The nominal interest rate is the three-month T-bill rate. The real interest rate is equal to the nominal rate minus the forecast of inflation by DRI, a private forecasting firm. The real exchange rate is the trade-weighted real exchange rate, normalized so that 1973 = 100. A negative trade surplus is a trade deficit.

# Pegs, Crawling Pegs

## 19.5 Fixed Exchange Rates

- We have assumed so far that the central bank chose the interest rate and let the exchange rate adjust freely in whatever manner was implied by equilibrium in the foreign exchange market.
- In some countries, this assumption does not reflect reality.
- Central banks act under implicit or explicit exchange rate targets and use monetary policy to achieve those targets.
- Countries that fix, or **peg**, their currencies must often intervene in the foreign exchange market to maintain their peg.
- The dedication to maintaining the peg varies. Some countries move slowly (i.e. **crawling peg**) while others move more rapidly.

# Bands, the EMS, and the Euro

## 19.5 Fixed Exchange Rates

- Another arrangement is for a group of countries to maintain their bilateral exchange rates within some **bands**.
- The most prominent example was the **European Monetary System (EMS)**, which determined the movements of exchange rates in the European Union from 1978 to 1998.
- Under the EMS rules, member countries agreed to maintain their exchange rate relative to other currencies in the system within narrow limits or bands around a **central parity**—a given value for the exchange rate.
- After a major crisis in 1992, which led several countries to drop out of the EMS, exchange rate adjustments became more and more infrequent, leading a number of countries to move one step further and adopt a common currency, the **euro**.

# Monetary Policy When the Exchange Rate Is Fixed

## 19.5 Fixed Exchange Rates

- Suppose a country decides to peg its exchange rate at some chosen value; call it  $\bar{E}$ .
- Let us look at the implications and mechanics of pegging.
- Recall that pegging or no pegging, the exchange rate and the nominal interest rate must satisfy the interest parity condition

$$(1 + i_t) = (1 + i_t^*) \left( \frac{E_t}{E_{t+1}^e} \right)$$

- When the country pegs its exchange rate at  $\bar{E}$ , the current exchange rate  $E_t = \bar{E}$  and  $E_{t+1}^e = \bar{E}$  as well.
- The interest parity relation becomes

$$(1 + i_t) = (1 + i_t^*) \Rightarrow i_t = i_t^*$$

# German Reunification, Interest Rates, and the EMS

## 19.5 Fixed Exchange Rates

- German reunification in 1990 led to large increases in demand and interest rate in Germany.
- To stay in the fixed exchange regime or the EMS, other European members had to match German interest rates, leading to decreases in their output.

| Table 1 Interest Rates, Inflation, and Output Growth after German Reunification: Germany, France, and Belgium, 1990–1992                                                                                                                  |                                   |      |      |                       |      |      |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|------|------|-----------------------|------|------|
|                                                                                                                                                                                                                                           | <i>Nominal Interest Rates (%)</i> |      |      | <i>Inflation (%)</i>  |      |      |
|                                                                                                                                                                                                                                           | 1990                              | 1991 | 1992 | 1990                  | 1991 | 1992 |
| Germany                                                                                                                                                                                                                                   | 8.5                               | 9.2  | 9.5  | 2.7                   | 3.7  | 4.7  |
| France                                                                                                                                                                                                                                    | 10.3                              | 9.6  | 10.3 | 2.9                   | 3.0  | 2.4  |
| Belgium                                                                                                                                                                                                                                   | 9.6                               | 9.4  | 9.4  | 2.9                   | 2.7  | 2.4  |
|                                                                                                                                                                                                                                           | <i>Real Interest Rates (%)</i>    |      |      | <i>GDP Growth (%)</i> |      |      |
|                                                                                                                                                                                                                                           | 1990                              | 1991 | 1992 | 1990                  | 1991 | 1992 |
| Germany                                                                                                                                                                                                                                   | 5.8                               | 5.5  | 4.8  | 5.7                   | 4.5  | 2.1  |
| France                                                                                                                                                                                                                                    | 7.4                               | 6.6  | 7.9  | 2.5                   | 0.7  | 1.4  |
| Belgium                                                                                                                                                                                                                                   | 6.7                               | 6.7  | 7.0  | 3.3                   | 2.1  | 0.8  |
| The nominal interest rate is the short-term nominal interest rate. The real interest rate is the realized real interest rate over the year—that is, the nominal interest rate minus actual inflation over the year. All rates are annual. |                                   |      |      |                       |      |      |
| Source: OECD Economic Outlook.                                                                                                                                                                                                            |                                   |      |      |                       |      |      |

# Fiscal Policy When the Exchange Rate Is Fixed

## 19.5 Fixed Exchange Rates

- If monetary policy can no longer be used under fixed exchange rates, what about fiscal policy?
- The effects of an increase in government spending when the central bank pegs the exchange rate are identical to those we saw in the slide 12 for the case of flexible exchange rates and an unchanged monetary policy.